

Sequences and Basic Series Convergence Tests



Convergence of sequences

- 1 Determine whether the sequence $a_n = \frac{3n+1}{n}$ converges, and if so, find its limit.
 - 2 Determine whether the sequence $a_n = (-1)^n$ converges.
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Geometric series

- 3 Determine whether $\sum_{n=0}^{\infty} 3\left(\frac{1}{2}\right)^n$ converges, and if so, find its sum.
 - 4 Determine whether $\sum_{n=1}^{\infty} 2^n$ converges.
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The n th-term (divergence) test

- 5 Use the n th-term test to show $\sum_{n=1}^{\infty} \frac{n}{n+1}$ diverges.
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The integral test

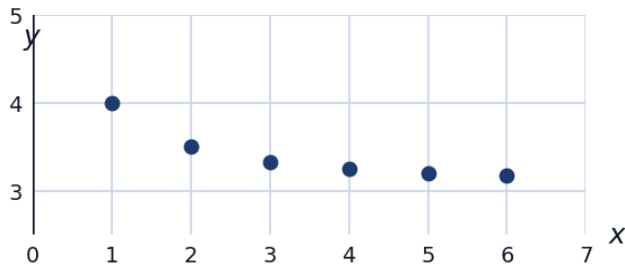
- 6 Use the Integral Test to determine whether $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges.
 - 7 Use the Integral Test to determine whether $\sum_{n=1}^{\infty} \frac{1}{n}$ converges (the harmonic series).
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The comparison and limit comparison tests

- 8 Use the Direct Comparison Test to determine whether $\sum_{n=1}^{\infty} \frac{1}{n^2+1}$ converges, comparing to $\sum \frac{1}{n^2}$.
 - 9 Use the Limit Comparison Test to determine whether $\sum_{n=1}^{\infty} \frac{2n}{n^2+3}$ converges, comparing to $\sum \frac{1}{n}$.
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Visualizing sequence convergence

- 10 Plot the first 6 terms of $a_n = \frac{3n+1}{n}$ (from the earlier question) and describe how the points behave as n increases.



Monotonic and bounded sequences

- 11 Determine whether $a_n = \frac{n}{n+1}$ is monotonic (always increasing or always decreasing), and find an upper bound.
- 12 State the Monotonic Sequence Theorem, and use it to conclude that $a_n = \frac{n}{n+1}$ converges.

The p-series test

- 13 State the p-series test for $\sum_{n=1}^{\infty} \frac{1}{n^p}$, and classify convergence for $p = 1$, $p = \frac{3}{2}$, and $p = \frac{1}{2}$.
- 14 Use the p-series test (after simplifying) to determine whether $\sum_{n=1}^{\infty} \frac{5}{n^{0.9}}$ converges.

Series review: choosing the right test

- 15 For each series, name the most efficient test to apply (do not fully evaluate):

a $\sum \frac{1}{5^n}$

b $\sum \frac{1}{n^3}$

c $\sum \frac{n+1}{2n+3}$

d $\sum \frac{1}{n^2 + 2n}$

Sequences defined recursively

- 16 A sequence is defined recursively by $a_1 = 2$, $a_{n+1} = \frac{a_n + 6}{2}$. Compute a_2 , a_3 , and a_4 , and conjecture the limit if the sequence converges.
- 17 Verify that if a recursively defined sequence $a_{n+1} = \frac{a_n + 6}{2}$ converges to a limit L , then L must satisfy $L = 6$.

Geometric series with algebraic manipulation

- 18 Find the sum of $\sum_{n=2}^{\infty} \left(\frac{2}{3}\right)^n$ (note the series starts at $n = 2$, not $n = 0$).
- 19 Express the repeating decimal $0.\overline{45}$ as a geometric series and find its exact fractional value.
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Telescoping series

- 20 Evaluate the telescoping series $\sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+1}\right)$ by examining its partial sums.